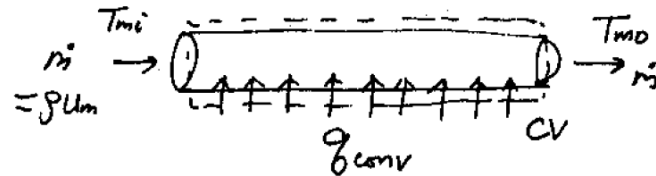
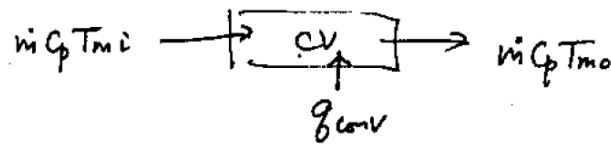


17. Pipe energy balance

[Overall energy balance for a pipe of length L]

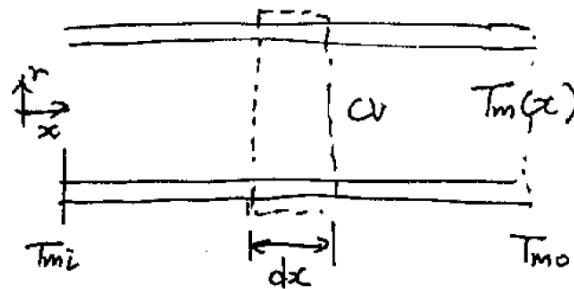


Energy balance (1st law of thermodynamics)



- $\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g$ or $q - \dot{W}_s = \dot{m}(h_e - h_i)$
- $0 = q_{conv} + \dot{m}c_p T_i - \dot{m}c_p T_o \rightarrow \boxed{q_{conv} = \dot{m}c_p (T_o - T_i)}$
- HT from pipe wall = $\dot{m}c_p \times \text{fluid temp. increase} = \text{Outlet enthalpy} - \text{Inlet enthalpy}$

[Mean temperature variation]



$$q_x = \dot{m}c_p T_m \rightarrow q_x + \frac{dq_x}{dx} dx = \dot{m}c_p T_m + d(\dot{m}c_p T_m)$$

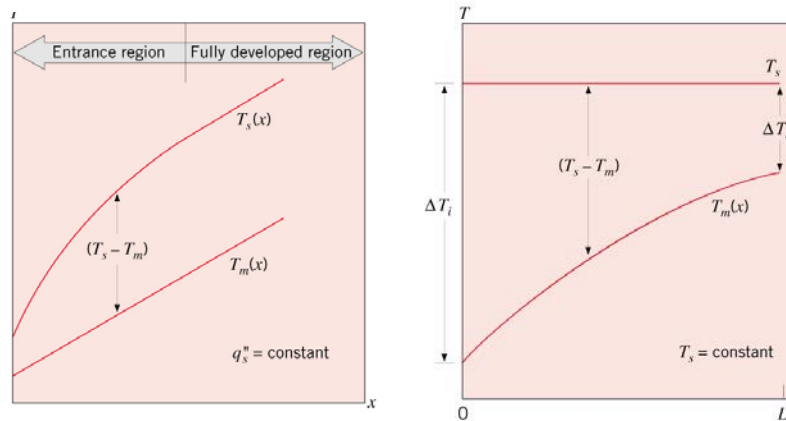
$$dq_{conv} = h(p dx) (T_s - T_m)$$

$$= q''_c p dx$$

- GE for mean temp. variation:

$$\frac{dT_m}{dx} = \frac{q''_s P}{\dot{m}c_p} \quad \text{or} \quad \frac{dT_m}{dx} = \frac{Ph}{\dot{m}c_p} (T_s - T_m) \quad (\text{perimeter } P = \pi D)$$

[Special cases]



(I) $q''_s = \text{const}$

$$\frac{dT_m}{dx} = \frac{q''_s P}{\dot{m} c_p} \rightarrow T_m(x) = T_{m,i} + \frac{q''_s P}{\dot{m} c_p} x$$

Linear increase of fluid temperature. Why?

(II) $T_s = \text{const}$

$$\frac{dT_m}{dx} = \frac{Ph}{\dot{m} c_p} (T_s - T_m) \rightarrow \frac{dT_m}{(T_s - T_m)} = \frac{Ph}{\dot{m} c_p} dx \rightarrow \int_{T_i}^{T_m} \frac{dT_m}{(T_s - T_m)} = \int_0^x \frac{Ph}{\dot{m} c_p} dx$$

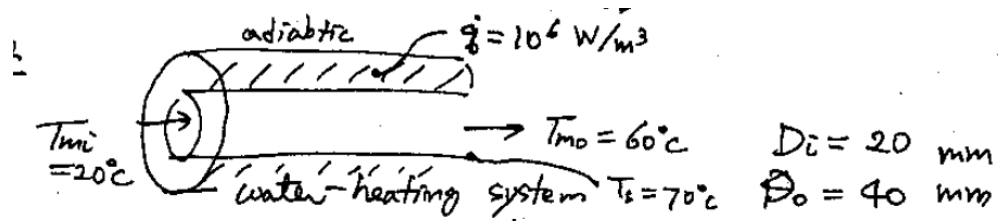
$$\rightarrow -\ln \frac{T_s - T_m}{T_s - T_i} = \frac{P}{\dot{m} c_p} \int_0^x h dx = \frac{Px \bar{h}_x}{\dot{m} c_p} \because \bar{h}_x = \frac{1}{x} \int_0^x h dx$$

$$\frac{T_s - T_m(x)}{T_s - T_{mi}} = \exp\left(-\frac{P \bar{h}_x}{\dot{m} c_p}\right)$$

Note that $\bar{h}_x$ is still unknown. Temp increase depends on HT coefficient.

Exponential increase of fluid temperature. Why?

(Example)



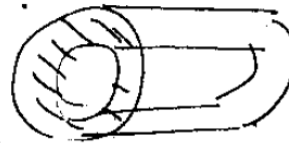
$\dot{m} = 0.1 \text{ kg/s}$ $L = ?$ $h_o = ?$

$\dot{E}_g = \dot{q}_{\text{conv}}$

$\dot{q}V = \dot{q} \pi (r_o^2 - r_i^2) L$

$= \dot{m} c_p (T_{mo} - T_{mi})$

all known except L



$L = 17.7$

$\dot{q}_s'' = h (T_s - T_m)$

$h_o = \frac{\dot{q}_s''}{T_s - T_{mo}}$

$\dot{q}_s'' = \frac{\dot{E}_g}{\pi D_i L} = \frac{\dot{q} \pi (r_o^2 - r_i^2)}{2\pi r_i L}$

$\therefore h_o = 1500 \text{ W/m}^2 \cdot \text{K}$